

Intelligent Agents

Chapter 2

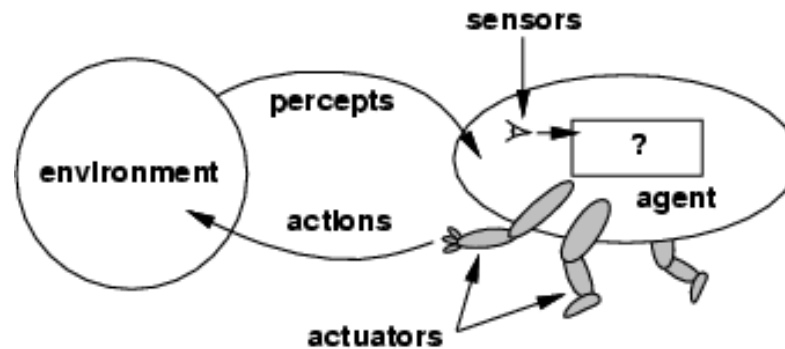
Outline

- Agents and environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types

Agents

- An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and **acting** upon that environment through **actuators**
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- Human agent: eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators
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- Robotic agent: cameras and infrared range finders for sensors; various arms/legs/motors for actuators
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Agents and environments



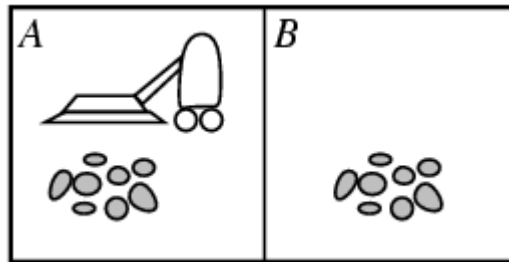
- The **agent function** maps from percept histories to actions :

$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$ (抽象的数学描述)

- The **agent program** runs on the physical **architecture** to produce f

agent = architecture + program (具体实现)

Vacuum-cleaner world



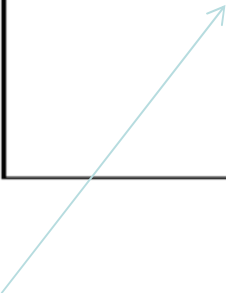
- Percepts: location and contents, e.g., [A,Dirty]
-
- Actions: *Left*, *Right*, *Suck*, *NoOp*
-

A vacuum-cleaner agent

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Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>

?



Rational agents

- An agent should strive to "do the right thing", based on what it can perceive and the actions it can perform. The right action is the one that will cause the agent to be most successful
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- Performance measure: An objective criterion for success of an agent's behavior
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- E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Rational agents

- **Rational Agent:** For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

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Rational agents

- Rationality is optimization under constraints. It's distinct from omniscience (all-knowing with infinite knowledge)
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- Agents can perform actions to gather useful information which could result in better performance (information gathering, exploration. 场景: 过马路)
-
- An agent is **autonomous** if its behavior is determined by its own experience (with ability to learn and adapt)
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Rational $\Rightarrow$ exploration, learning, autonomy

PEAS

- PEAS: Performance measure, Environment, Actuators, Sensors (**task environment**)
- Must first specify the setting of these four aspects before the design of an intelligent agent
- Consider, e.g., the task of designing an automated taxi driver:
- - Performance measure?
 - Environment ?
 - Actuators ?
 - Sensors?

PEAS – automated taxi driver

- Agent: an automated taxi driver
- Performance measure: Safe, fast, legal, comfortable trip, maximize profits
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- Environment: Roads, other traffic, pedestrians, customers, weather
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- Actuators: Steering wheel, accelerator, brake, signal, horn, display
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- Sensors: Cameras, sonar, speedometer, GPS, odometer, engine sensors, gauges, keyboard
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PEAS - Medical diagnosis system

- Agent: Medical diagnosis system
- Performance measure: Healthy patient, minimize costs and lawsuits
- Environment: Patient, hospital, staff
- Actuators: Screen display (questions, tests, diagnoses, treatments, referrals)
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- Sensors: Keyboard, mouse (entry of symptoms, findings, patient's answers)

PEAS - Part-picking robot

- Agent: Part-picking robot
- Performance measure: Percentage of parts in correct bins
- Environment: Conveyor belt with parts, bins
- Actuators: Jointed arm and hand
- Sensors: Camera, joint angle sensors

PEAS - Interactive English tutor

- Agent: Interactive English tutor
- Performance measure: Maximize student's score on test
- Environment: Set of students
- Actuators: Screen display (exercises, suggestions, corrections)
- Sensors: Keyboard, mouse

Environment types

- **Fully observable** (vs. partially observable): An agent's sensors give it access to the complete state of the environment at each point in time.
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- **Deterministic** (vs. stochastic): The next state of the environment is completely determined by the current state and the action executed by the agent. (If the environment is deterministic except for the actions of other agents, then the environment is **strategic**)
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- **Episodic** (vs. sequential): The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action), and the choice of action in each episode depends only on the episode itself.

Environment types

- **Static** (vs. dynamic): The environment is unchanged while an agent is deliberating. (The environment is **semidynamic** if the environment itself does not change with the passage of time but the agent's performance score does)
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- **Discrete** (vs. continuous): A limited number of distinct, clearly defined percepts and actions.
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- **Single agent** (vs. multiagent): An agent operating by itself in an environment.

Environment types

	Chess with a clock	Chess without a clock	Taxi driving
Fully observable	Yes	Yes	No
Deterministic	Strategic	Strategic	No
Episodic	No	No	No
Static	Semi	Yes	No
Discrete	Yes	Yes	No
Single agent	No	No	No

- 打牌的环境？
- The environment type largely determines the agent design
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- The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent
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Agent functions and programs

- An agent is completely specified by the agent function mapping percept sequences to actions
- Our aim: find a way to implement the rational agent function concisely
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- 至于把函数变成程序，不做具体讨论

Table-lookup agent

- Function `TABLE_DRIVEN_AGENT(percept)` returns an action
 - append precept to percepts;
 - action = `LOOKUP (percepts, table);`
 - return action
- Drawbacks:
 - Huge table
 - Take a long time to build the table
 - No autonomy
 - Even with learning, need a long time to learn the table entries

例子：求平方根

- 1.表（我们学对数的时候还用对数表）
- 2.牛顿法(find b's square root x)

$$\text{New } x = \frac{1}{2} \left(x + \frac{b}{x} \right)$$

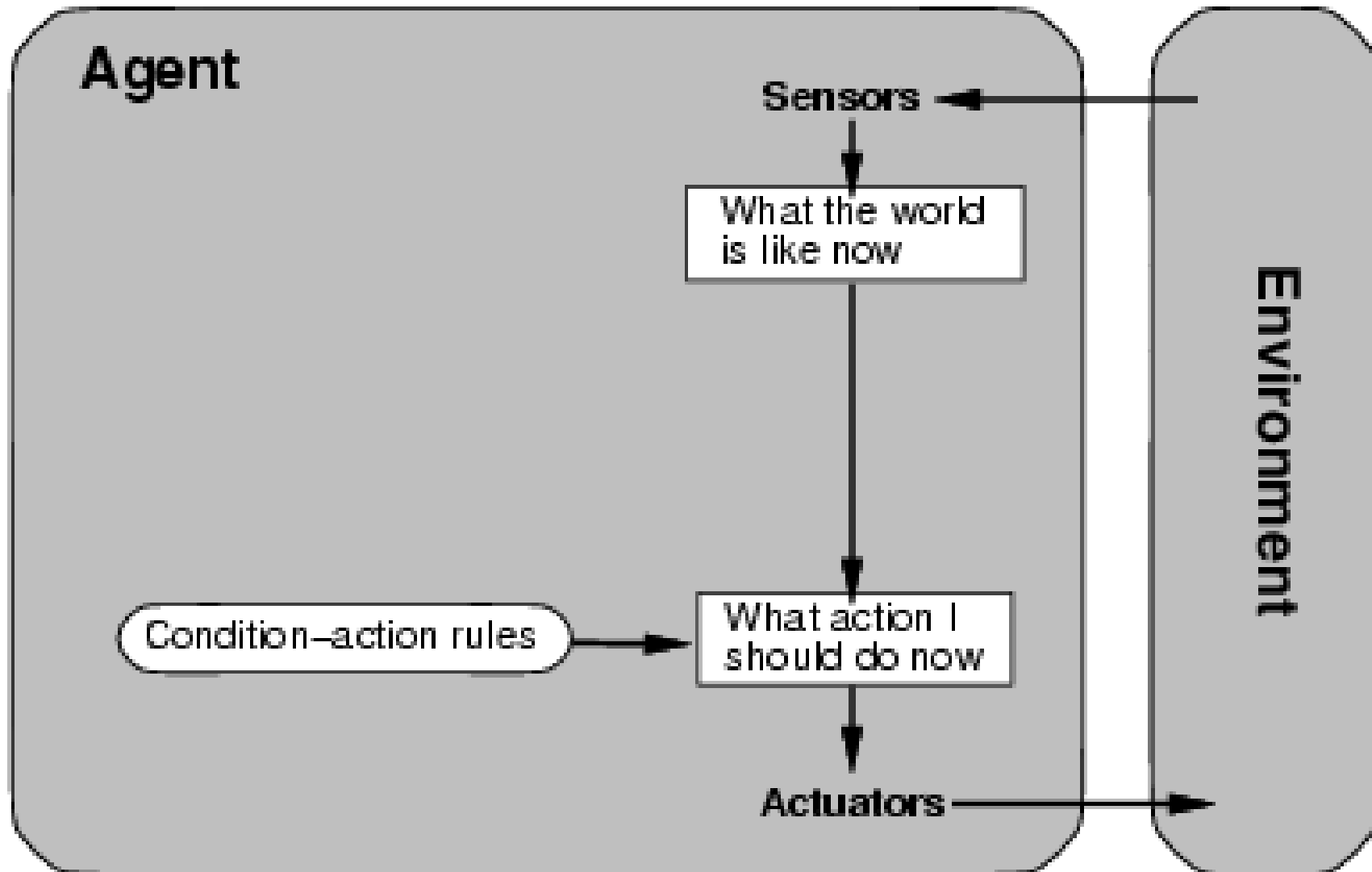
Agent program for a vacuum-cleaner agent

```
function REFLEX-VACUUM-AGENT( [location,status] ) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

Agent types

- Four basic types in order of increasing generality:
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 - Simple reflex agents
 - Model-based reflex agents
 - Goal-based agents
 - Utility-based agents
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- All these can be turned into learning agents

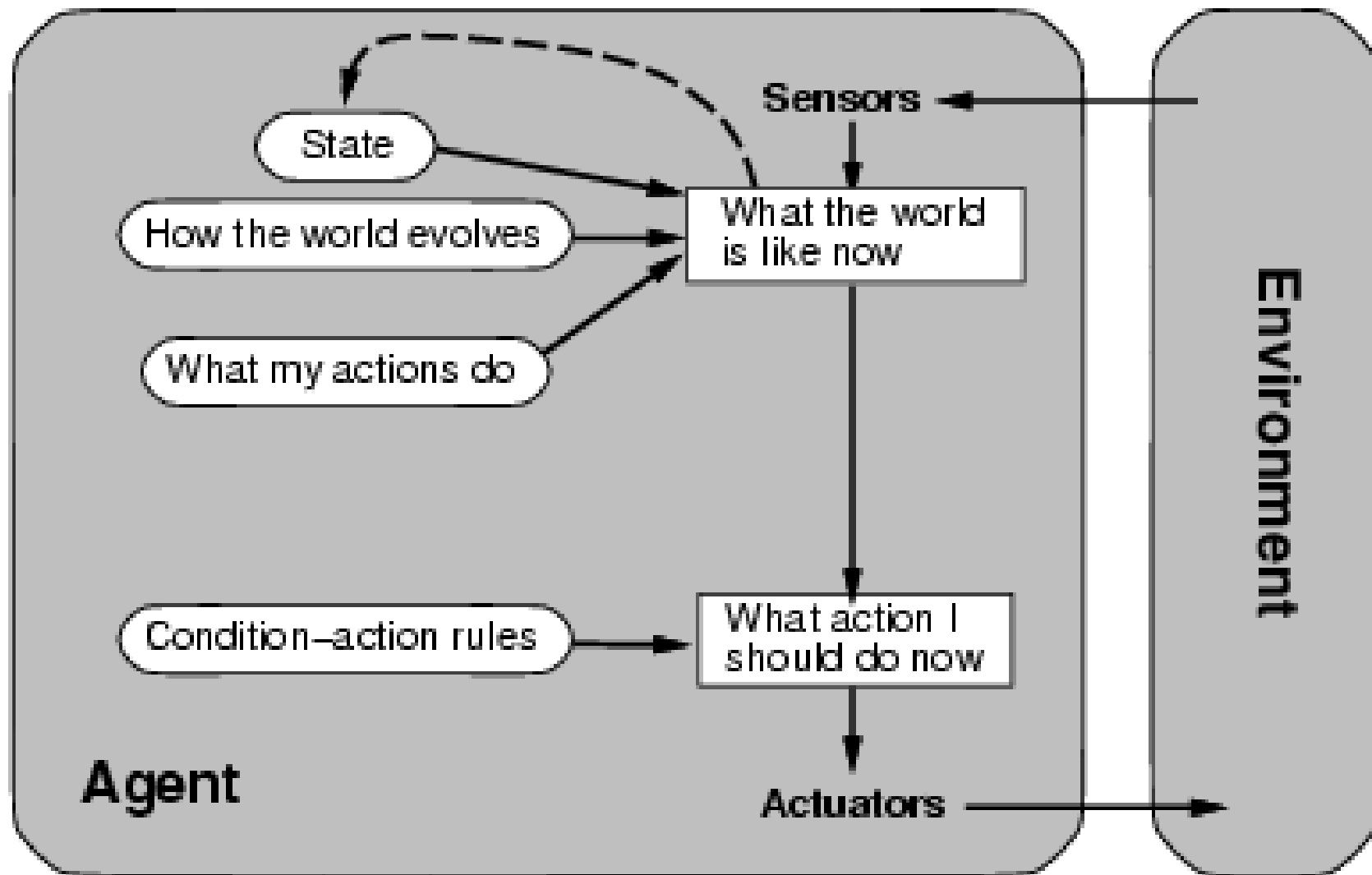
Simple reflex agents



Simple reflex agents

- Function `SIMPLE_REFLEX_AGENT(percept)` returns an action
- - `state = INTERPRET-INPUT(percept);`
 - `rule = RULE-MATCH(state, rules);`
 - `action = RULE-ACTION[rule];`
 - `return action`
- Acts on the *current* percept / fully observable environment

Model-based reflex agents



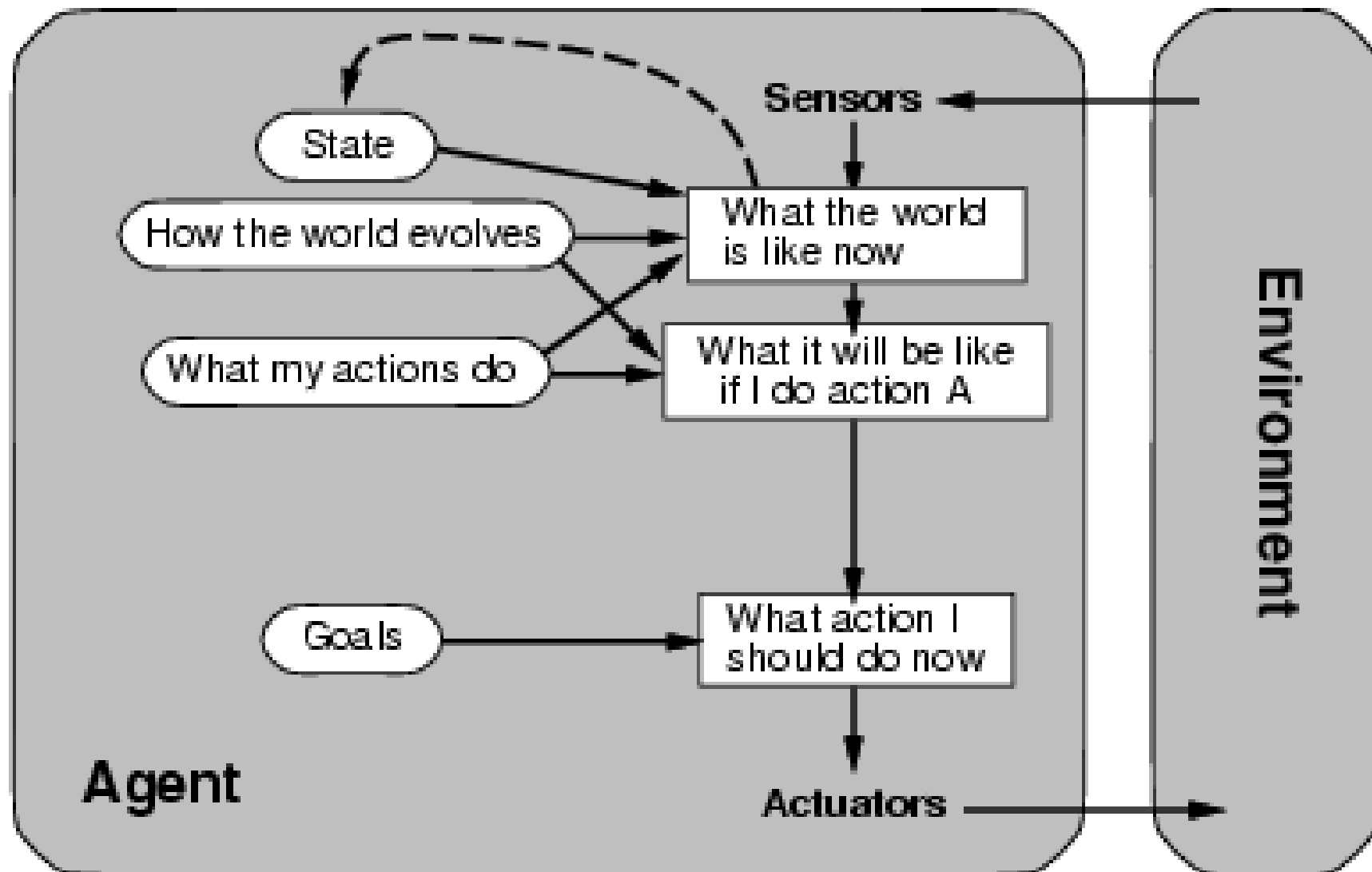
模型=对世界的状态描述+世界如何演变
我的行动的效果可影响世界 (=环境+我)

Model-based reflex agents

- Function
REFLEX_AGENT_WITH_STATE(percept)
returns an action
- ```
state = UPDATE-STATE(state,action,percept);
rule = RULE-MATCH(state, rules);
action = RULE-ACTION[rule];
return action
```

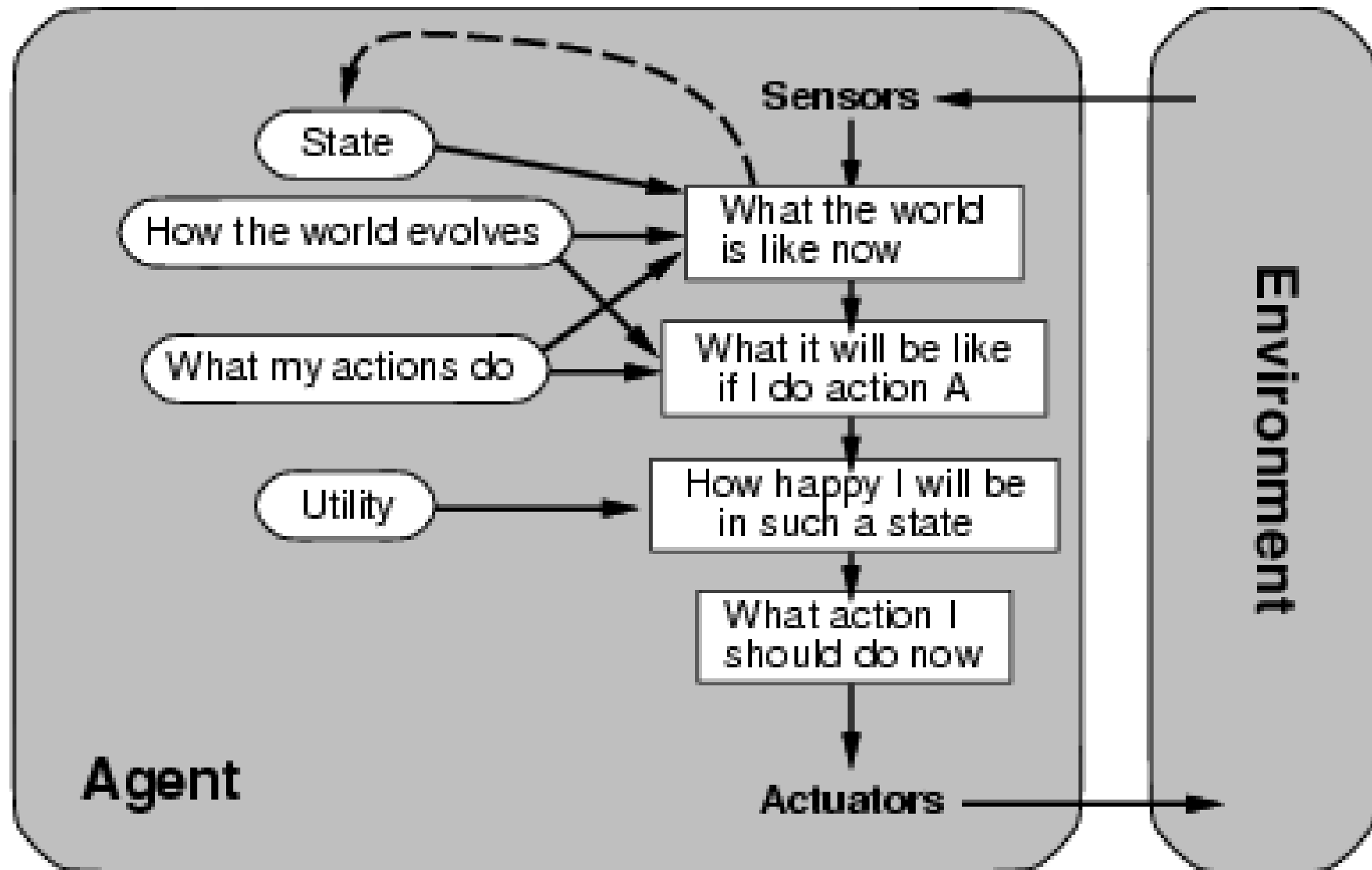
最简单的模型是有限状态机。

# Goal-based agents

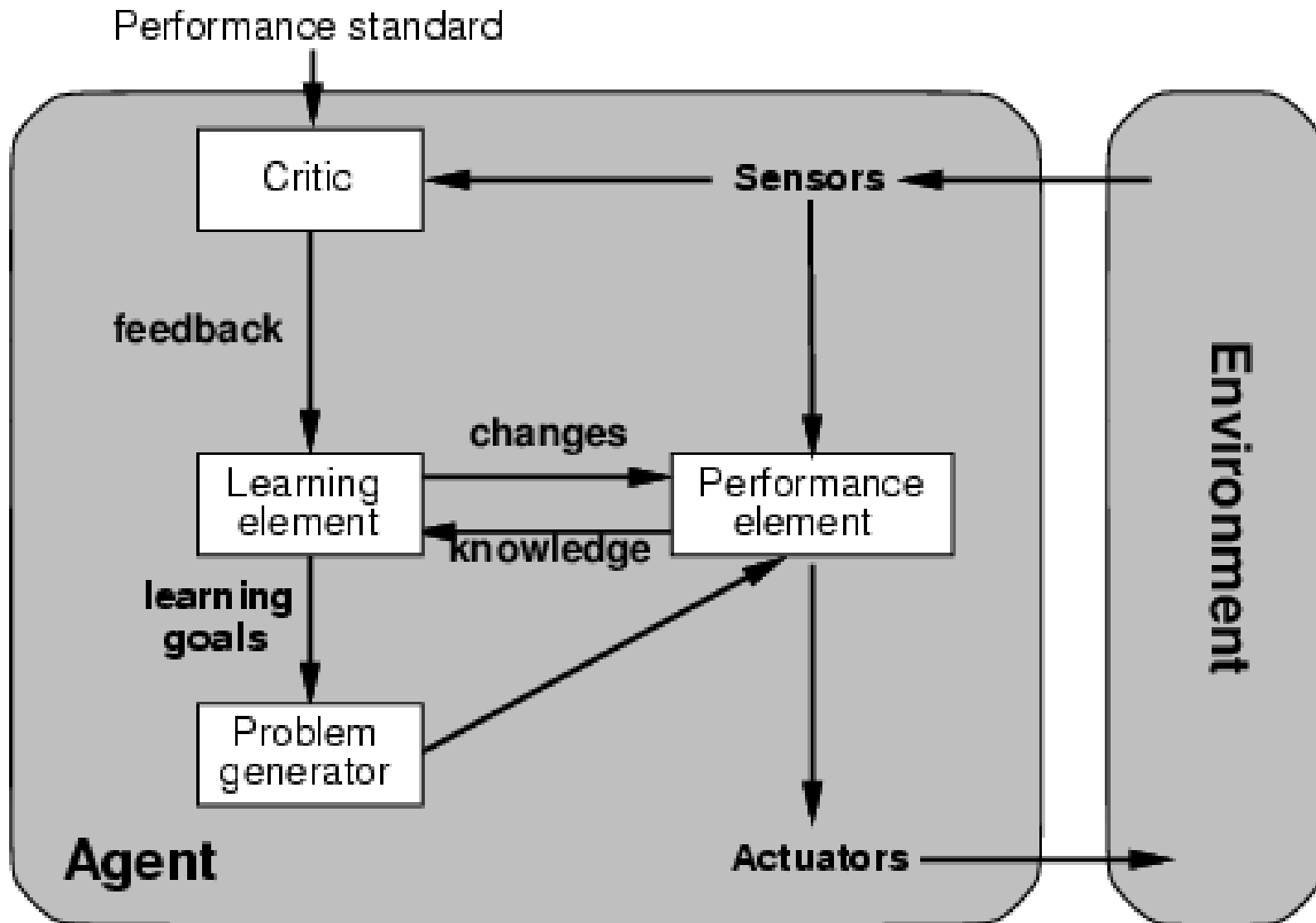


- We need *search* and *planning* to achieve the goal
- 搜索是下一章的主题

# Utility-based agents



# Learning agents



# 具有学习能力的agent

- 什么是（机器）学习？
  - [An agent] is said to learn from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ , if its performance at tasks in  $T$ , as measured by  $P$ , improves with  $E$ . (Mitchell, 1997, p. 2)
- 要素
  - Experience(data/corpus),
  - Critic,
  - Target function (task, performance measure),
  - Algorithms (knowledge)

# 具有学习能力的agent

- 数据是学习的核心要素
  - 学习不是演绎推理，是归纳推理。处理新数据是关键
- **Critic**确定了学习的分类
  - 有监督学习
  - 无监督学习
  - 半监督学习,弱监督学习,强化学习
- 目标函数决定了学习的难易程度
  - 一般，目标函数是已知的，学习算法只是调整目标函数的参数

# 小结

Agents interact with environments through actuators and sensors

The agent function describes what the agent does in all circumstances

The performance measure evaluates the environment sequence

A perfectly rational agent maximizes expected performance

Agent programs implement (some) agent functions

PEAS descriptions define task environments

Environments are categorized along several dimensions:

observable? deterministic? episodic? static? discrete? single-agent?

Several basic agent architectures exist:

reflex, reflex with state, goal-based, utility-based

# 作业

- 2.5 For each of the following agents, develop a PEAS description of the task environment:
  - a. Robot soccer player;
  - b. Internet book-shopping agent;
  - c. Autonomous Mars rover;
  - d. Mathematician's theorem-proving assistant.